EES 102: Introduction to Environmental Sciences

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Image: https://wiki.seg.org/index.php?curid=13659

THE BIG ENERGY STEPS

1. Agriculture diverts more sunlight captured through photosynthesis to humans
2. Domestication of animals increases our “horse power” from 75W to 700W
3. Deforestation, irrigation and drainage of wetlands allowed us to take over new ecosystems and divert them towards agriculture
4. The industrial revolution provided energy from plant material that got preserved in certain geological settings millennia ago (coal, oil and gas)
5. Renewable energy (wind, solar, radiative cooling)

Renewable energy comprises of a number of techniques to capture the energy of the sun more directly without going through the process of photosynthesis. Photosynthesis captures only 1-2% of the incident energy so it is easy to see that a solar panel with 20-40% efficiency requires much less space. Novel materials can also help us to dispose excess heat (>100 W//m^2) directly to space through the atmospheric window. This allows us to get more energy from less land and consume less for cooling. This can lead us into an era where the equation: increased energy consumption = increasing destruction of natural ecosystems

need not hold any longer. More efficient & less destructive energy harvest can get us out of the Anthropocene. Anyway fossil fuels are finite - no alternatives!

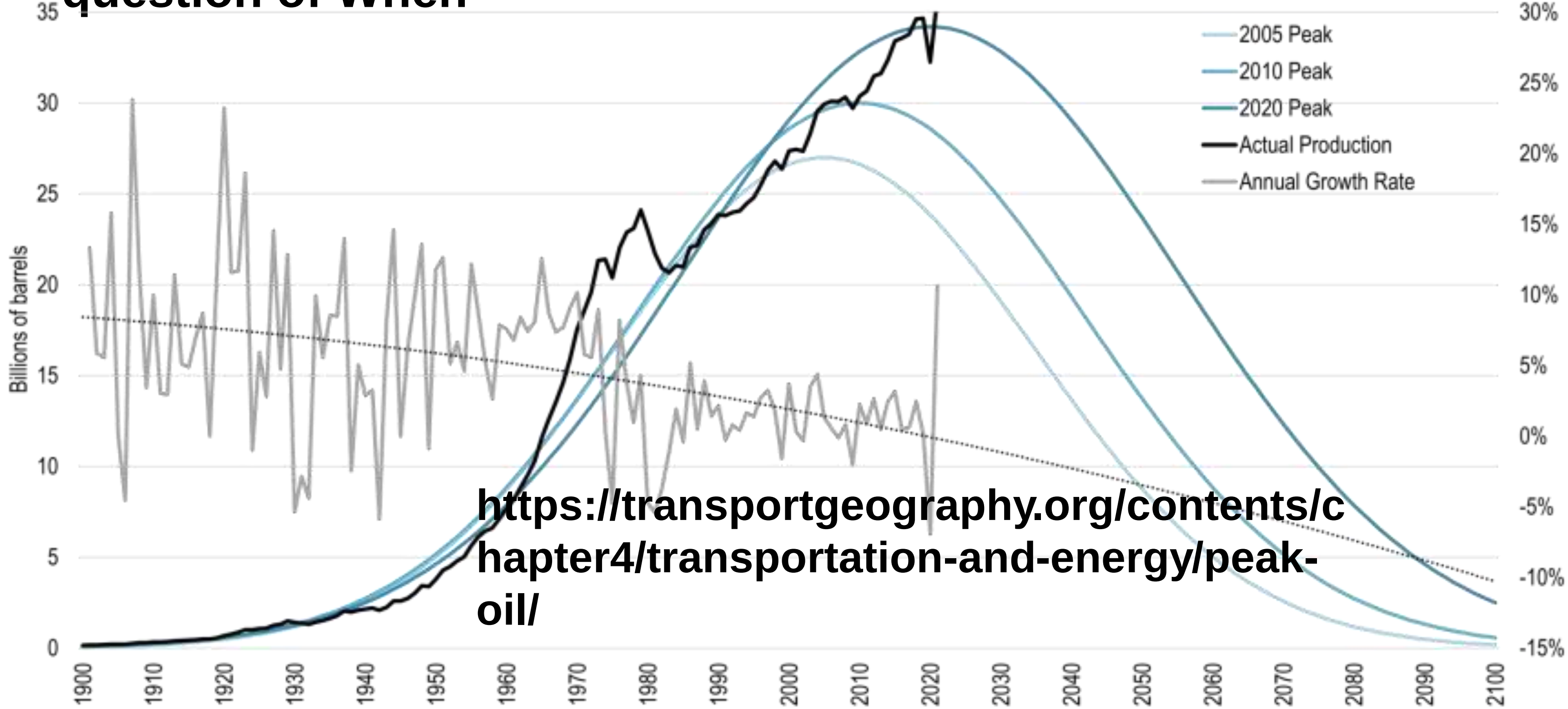
HOW DO WE GET OUT OF THE ANTHROPOCENE?

- We humans are remarkably adaptable, innovative, and good at cooperation.
- Most women do not aspire to raise large families – ensuring everyone has only as many children as they want will result in shrinking populations
- Scientific findings, emerging technologies, innovations, and existing know-how allow us to embark on the most challenging and exciting journey ever. A transition to a world that reconnects our society and well-being to the biosphere, and generates human prosperity within safe planetary boundaries with renewable energies and agro technologies that permit to grow more food per m^2 land and litre of water which can permit us to re-wilder large territories.
- This journey requires focus on human happiness and well being and on equitable growth.
- A thriving global society, now and in the future, is impossible without living forests, lakes, waterfalls and oceans. We cannot live without and atmosphere, and bio-geochemical cycles that keep our climate stable and without an ozone layer that protects us from harmful UV radiation.

PEAK OIL: UNDERSTANDING THE ISSUE

- **Peak oil is the point in time when the maximum rate of global oil production is reached, after which production will begin an irreversible decline.**

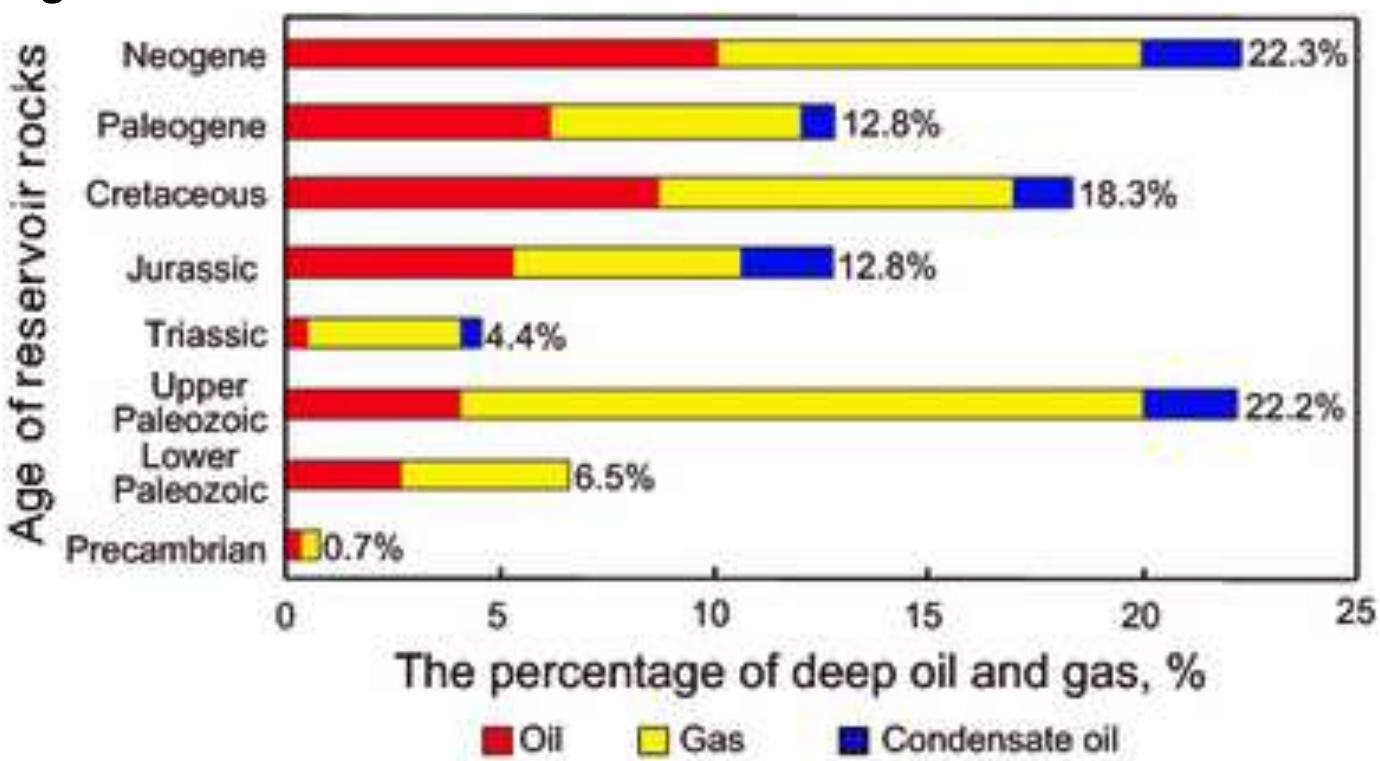
- **Not really a question of *if* : Probability 100%, Only a question of When**



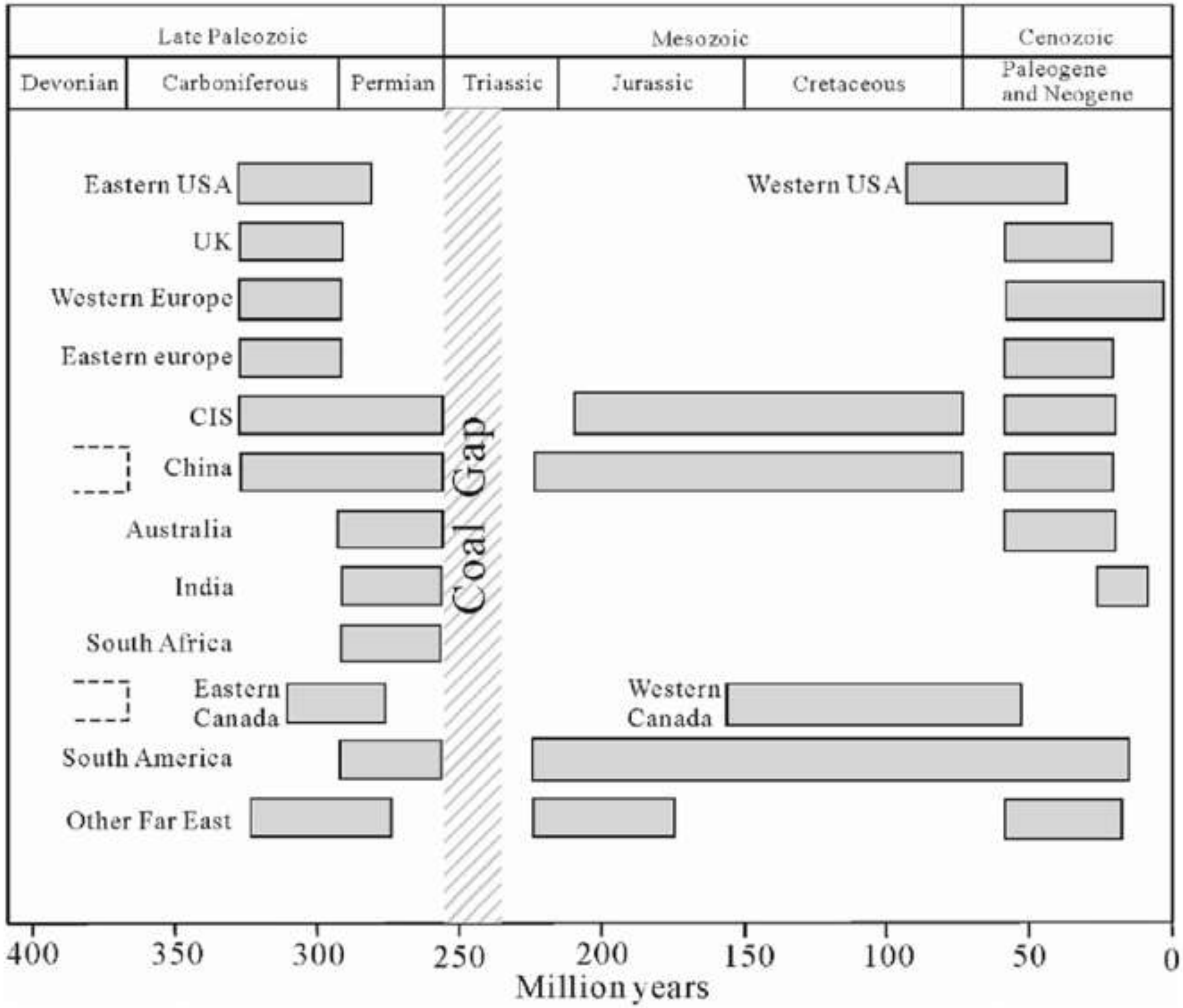
https://transportgeography.org/contents/chapter4/transportation-and-energy/peak-oil/

PEAK OIL: THE AGE OF WORLD OIL

Fossil fuels contain old plants and marine organisms. The carbon was formed through the process of photosynthesis years ago. The supply is finite and the resource is not reformed at the rate at which we are consuming it.



THE AGE OF THE WORLDS COAL

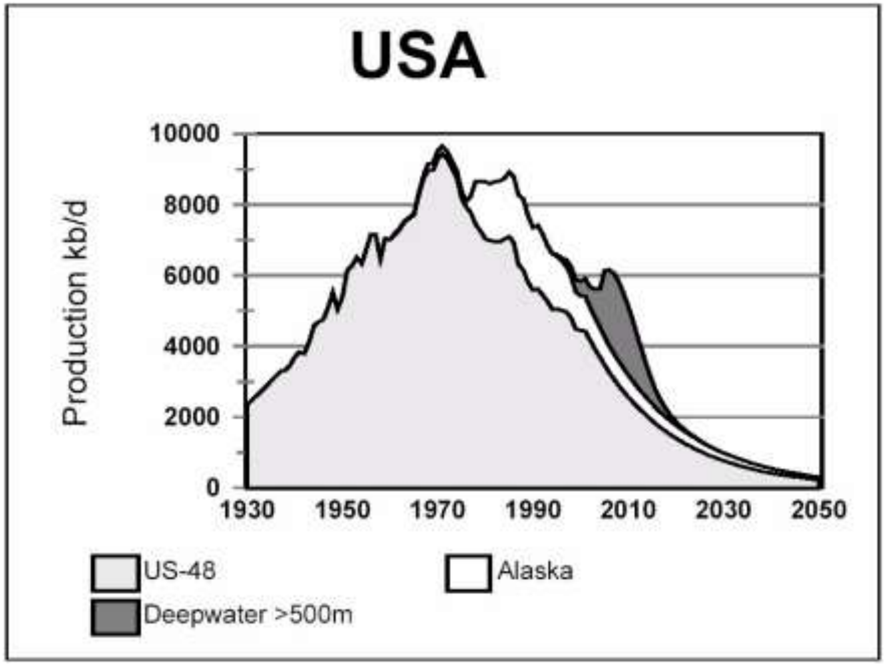


PEAK OIL DISCOVERER: DR. KING HUBBERT 1903-1989

- Shell Oil Geologist/ Petroleum Scientist
- 1949 – projected short historical oil period
  - Triggered by 1930 U.S. discovery peak
- 1956 – predicted 1970 as U.S. Peak Oil year
  - Came as predicted
- 1969 – predicted World Peak Oil year 2000
  - 1970-80 demand decline delayed it

OIL DISCOVERY/PRODUCTION PEAKS – U.S.

- **1930 – U.S. "lower 48" Oil <u>Discovery</u> Peak year**
- **1970 – U.S. "lower 48" Oil <u>Production</u> Peak year**
- **Peak discovery/peak production lag time of 40 years**



DR. COLIN CAMPBELL

- Geologist/ Petroleum Scientist

- Worked for most major oil companies

- Founder, Association for Study of Peak Oil
  - Wrote “The Coming Oil Crisis” in 1997

- Estimates World Peak for regular oil in 2010

- Published two other books
  - “Essence of Oil and Gas Depletion”
  - “Oil Crisis”

MATTHEW SIMMONS

- Oil Investment Banker
  - Backed many oil and gas drilling projects

- Advisor to President Bush

- Challenged Saudi Reserve Estimates
  - Thought Saudi oil may soon peak

- Author, “Twilight in the Desert: The Coming Saudi Oil Shock and the World Economy”

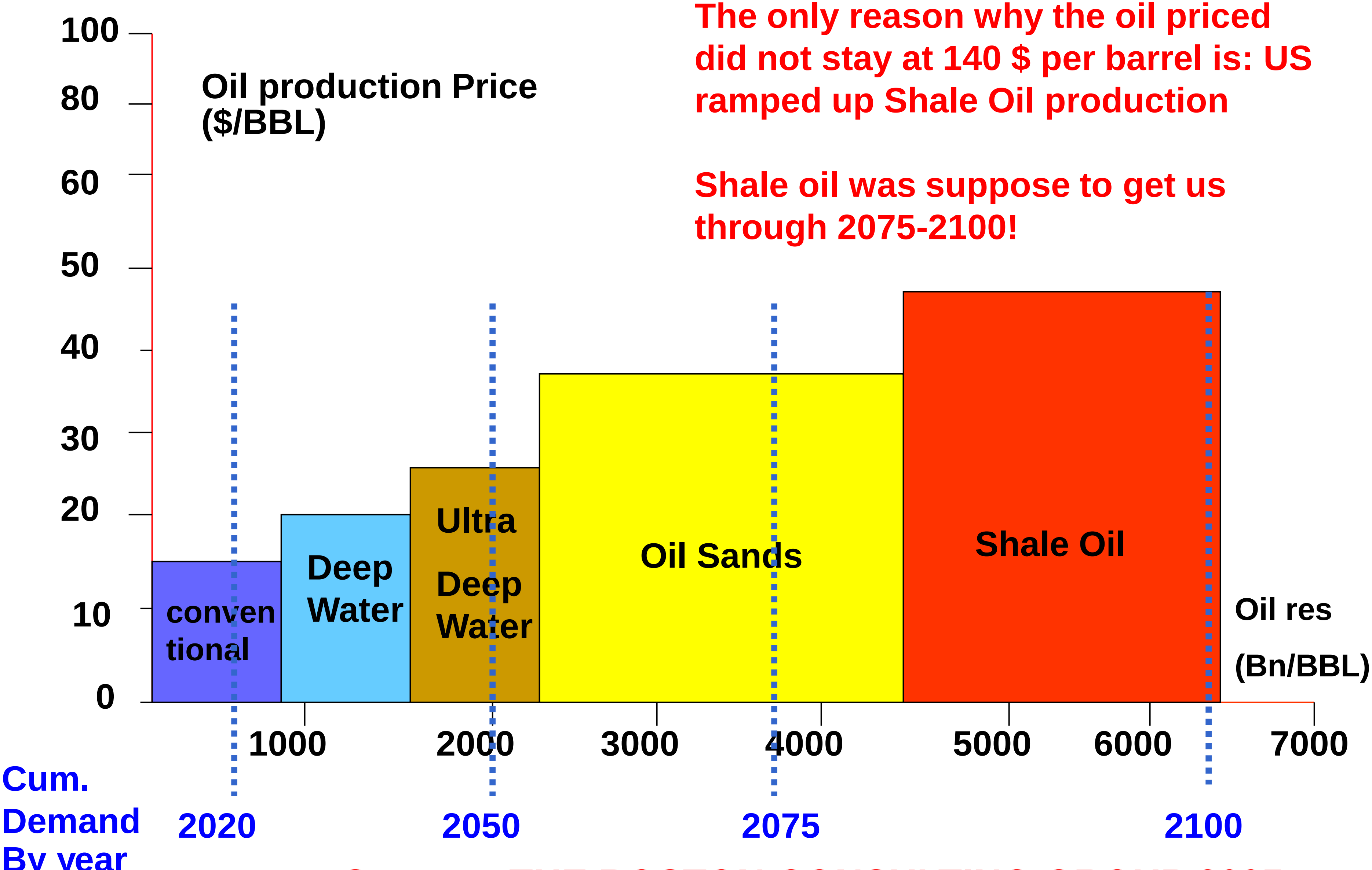
- Given 100s of talks to government and business

The support staff of the US military surely includes some excellent geophysicists… which is why shortly after the second gulf war US went all out to develop shale oil and become independent of the global oil market. 2012 prices were >100$/barrel. The crash in prices after 2014 is thanks to shale oil

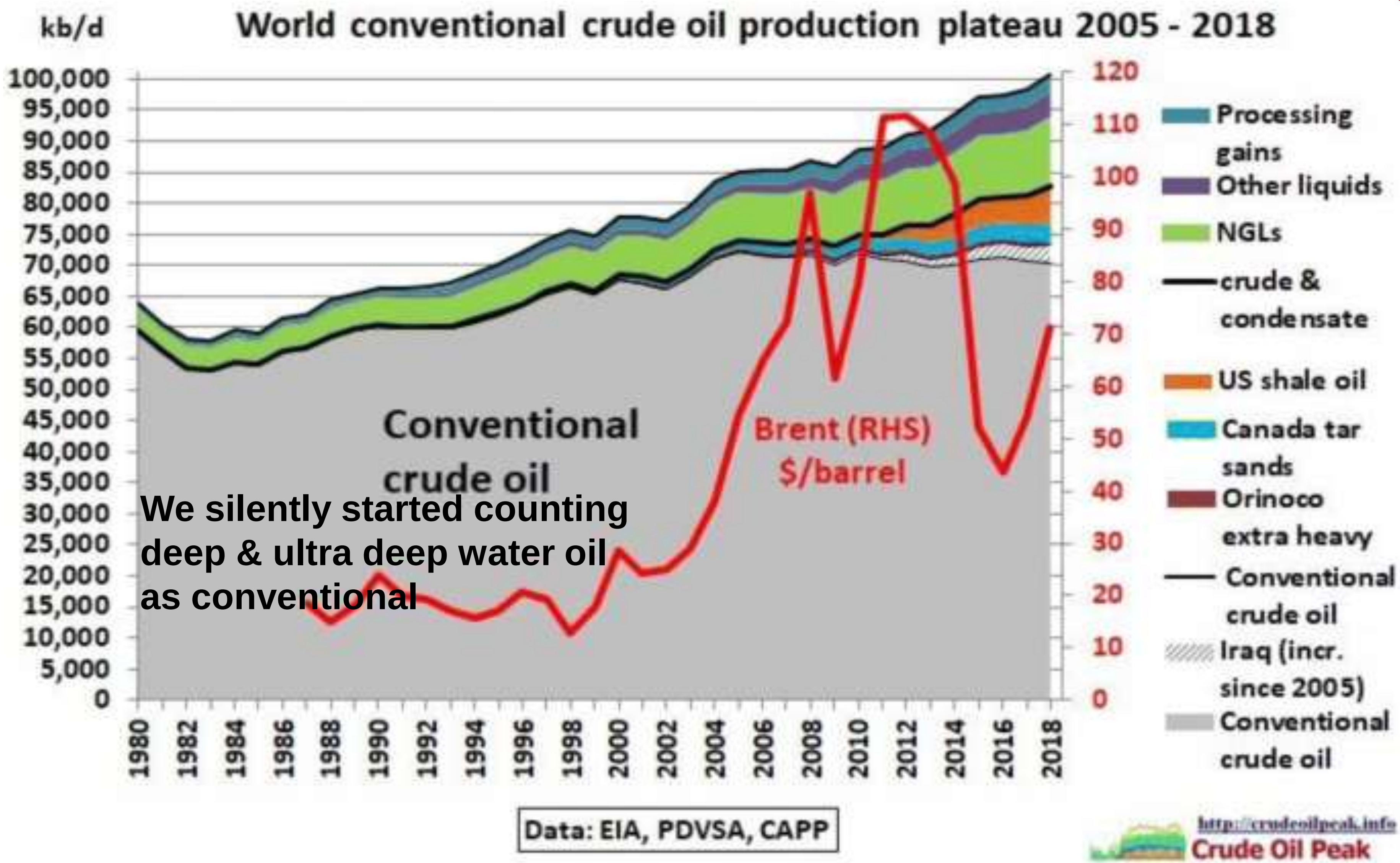
During the 2nd gulf war US government started to heavily supported the only oil left in the USA: shale oil

The only reason why the oil priced did not stay at 140 $ per barrel is: US ramped up Shale Oil production

Shale oil was suppose to get us through 2075-2100!



Source : THE BOSTON CONSULTING GROUP 2005



We silently started counting deep & ultra deep water oil as conventional

CHANGING OUR TRANSPORT SECTOR IS URGENT

Conventional oil peaked in ~2010 as Dr. Campell (Nickname: King Hubbert II) predicted

Of ~ 100 mbd current production approximately
~10-15 mbd are deep & ultra deep water
~10 mbd shale oil
~ 5 mbd oil sands

We need to worry more about how we will fly airplanes in 30 years and less about the political backlash of banning petrol/diesel fuelled 2W & 3W

Conventional oil did not get us till 2025
No reason to assume shale oil will get us to 2100

Some Important Concepts

- **The concept of Natural Capital (Paul Hawken): Natural resources are Air, Water, Soil, Energy and Minerals**
- **Difference between preservationists, conservationists and restorationists**
- **Most present day economic models do not factor in “accurate accounting” e.g. cost to environment**
- **GDP: Gross domestic product: The market value of in current dollars of all goods and services produced (and TAXED) within a country during a year**
- **Presently, 1 % of world population has 44% of the world wealth (assets) and income, generates 50% of the GHG emissions use 48% of its natural resources and generate 35% of its waste. EOD of the top 1% =Jan 10th**

Perpetual, Renewable and Non Renewable Resources

- **Perpetual resource: e.g. solar , wind , flowing water: On human time scale, these resources are renewed continuously (Sun's life: 6 billion years)**
- **Renewable resource: Resources that can be replenished fairly rapidly (hours to several decades) through natural processes as long as it is not used up faster than it is replaced e.g. forests, grasslands, wild animals, fresh air**
- **Non-renewable resources: These resources do not get replenished on human time scales**
- **The "tragedy of commons" 1968 Biologist Garret Hardin**

Typical responses of people to environmental problems

1) **Denial : Find someone or something to blame :** Its them not us (the industry, politicians, environmentalists) They are the villains and we are the victims

   However we may realize we make direct and indirect contributions to it too!

2) **Gloom and doom pessimism** (it is hopeless!)
3) **Blind technological optimism** (science and technofixes will save us)
4) **Fatalism** (We have no control over our actions and the future)
5) **Extrapolation to infinity**
6) **Paralysis by analysis** (seek perfection before doing anything)
7) **Faith in simple , easy answers**

   We should aim to keep hope above despair